

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Code of Conduct:

I G. Panthulu Kumar (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Panthulu Kumar

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Understanding	20%				
Method Selection	20%				
Solution Process	25%				
Accuracy of Calculation	20%				
Interpretation of Results	15%				

Total Marks **/40**

Signature of the Faculty

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on

the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method	20%	Selects most	Selects	Inappropriate	Unable to

Selection		appropriate method/formula with strong justification	correct method with minor errors	or partially correct method	
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Co2-AT1
 Subnetive
 Given

Q2-AT1 - Analytical problem

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CSA-0707

1) Subnetting of 192.168.20.0/24

Given network

192.168.20.0/24

Lab A (126)

- Subnet Mask: 255.255.255.192
- Block Size = 64

Subnet:

- Network address: 192.168.20.0
- First Host: 192.168.20.1
- Last Host: 192.168.20.62
- Broadcast address: 192.168.20.63

Usable hosts:

$$2^{(32-26)} - 2 = 2^6 - 2 = 64 - 2 = 62$$

Lab B (127)

Next available subnet starts from 192.168.20.64

Subnet Mask:

255.255.255.224

- Network address: 192.168.20.64
- First host: 192.168.20.65
- Last host: 192.168.20.94
- Broadcast: 192.168.20.95

Usable Hosts

$$2^5 - 2 = 32 - 2 = 30$$

LAB C (128)

next available subnet starts from 192.168.20.96

Subnet Mask:

255.255.255.240

- network address: 192.168.20.96
- first host: 192.168.20.97
- last host: 192.168.20.110
- broadcast: 192.168.20.111

Usable Hosts

$$2^4 - 2 = 16 - 2 = 14$$

Q. Systems using 126 Mask

Subnet Mask

255.255.255.192

Block size = 64

Subnets are

• 0-63

• 64-127

• 128-191

• 192-255

System 2

IP:

192.168.1.130

falls in:

128-191

- Network: 192.168.1.128
- First host: 192.168.1.129
- Last host: 192.168.1.190
- Broadcast: 192.168.1.191

Yes, 192.168.1.130 belongs to this subnet.

System 2

IP:

192.168.1.190

Also falls in

128-191

- Network: 192.168.1.128
- First Host: 192.168.1.129
- Last host: 192.168.1.190
- Broadcast: 192.168.1.191

Yes, 192.168.1.190 belongs to this subnet.

usable hosts

$$2^6 - 2 = 62$$

3. Organization Network

172.16.0.0/16

Assign Subnets Sequentially

HR (125)

- Network: 172.16.0.0
- Host: 172.16.0.1 - 172.16.0.126
- Broadcast: 172.16.0.127

Hosts

$$2^7 - 2 = 126$$

Sales (126)

Next subnet

172.16.0.128

- Host: 172.16.0.129 - 172.16.0.190
- Broadcast: 172.16.0.191

Hosts

30

IT (127)

Next subnet

172.16.0.192

- Host: 172.16.0.193 - 172.16.0.222
- Broadcast: 172.16.0.223

Hosts
30

30

$$2^5 - 2 = 30$$

admin (128)

next subnet

172.16.0.224

• Host: 172.16.0.225 - 172.16.0.238

• Broadcast: 172.16.0.239

Hosts

$$2^4 - 2 = 14$$

Remaining unused Range

Allocated:

172.16.0.0 - 172.16.0.239

Remaining:

172.16.0.240 - 172.16.255.255

4. Company Network

network

172.16.10.0/24

Subnet Mask

/27

Total Subnets

$$2^{27-24} = 2^3 = 8$$

Hosts per Subnet

$$2^5 - 2 = 30$$

Subnet List

<u>network</u>	<u>Range</u>	<u>Broadcast</u>
0	0-31	31
32	32-63	63
64	64-95	95
96	96-127	127
128	128-159	159
160	160-191	191
192	192-223	223
224	224-255	255

IP 172.16.10.78

falls inside

64-95

Network:

172.16.10.64

Host-Range:

65-94

cast:
95

IP 172.16.10.95

95 is the broadcast address of the subnet.
therefore,

NO. It is not a valid host address, although it belongs to the same subnet as the broadcast address.

5. IPV6

Given

2001:0db8:0000:0000:0000:ff00:0042:8329

Compressed form

Remove leading zeros.

Replace longest consecutive 0000 groups with ::

2001:db8::ff00:42:8329

Expanded form

2001:0db8:0000:0000:0000:ff00:0042:8329

Network prefix (/64)

first 64 bits remain.

2001:0db8:0000:0000::/64

or)

2001:db8:0:0::/64

Number of host address

host - 1678

$$128 - 64 = 64$$

$$2^{64} = 18,446,744,073,709,551,616 \text{ possible address}$$